

Dharun Muthaiah Nataraj

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EDUCATION

Carnegie Mellon University

M.S. in Civil and Computer Engineering (AI & ML Systems Focus), GPA: 4.0/4.0

Pittsburgh, PA

Dec 2026

Coursework: Intro to Deep Learning, Visual Learning and Recognition, LLM Systems, ML for Radars

Vellore Institute of Technology

B.Tech in Electronics and Communication Engineering, GPA: 9.18/10

Chennai, India

May 2025

SKILLS

Languages: Python, C++, MATLAB

ML Systems: Transformers, LLMs, VLMs, vLLM, FlashAttention, Quantization, LoRA, Speculative Decoding, Distributed Training

ML/Data: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV, YOLO

Systems/Tools: Triton, CUDA, Docker, AWS, MongoDB, FastAPI, Git, Linux, PyTorch Profiler, Nsight Compute

EXPERIENCE

NoRILLA | Research Intern (ASR Systems)

Jun 2026 – Present

Tech: Whisper, Parakeet, Canary-Qwen, NeMo, LoRA

Pittsburgh, PA

- Benchmarked CPU inference latency, throughput, and transcription accuracy across ASR models (Whisper, Qwen-3.5, Parakeet, Canary-Qwen) to evaluate deployment trade-offs for on-device speech recognition.
- Fine-tuned ASR models using LoRA and NVIDIA NeMo, optimizing data preprocessing and training pipelines to improve experimentation efficiency on NVIDIA A100 GPUs.

Carnegie Mellon University | Research Intern (ML Systems)

Jan 2026 – Apr 2026

Advisor: Prof. Tze Meng Low | Tech: Triton, PyTorch, PyTorch Profiler, NVIDIA Nsight Compute

Pittsburgh, PA

- Ported the SmaLL framework for machine learning libraries from CPU to GPU, implementing optimized Triton kernels that achieved a 1.87x speedup over PyTorch/cuDNN.
- Profiled GPU kernels using PyTorch Profiler and analyzed generated PTX, optimizing memory access, tiling, and kernel scheduling for convolution and reduction kernels.

MacV AI | Computer Vision Engineer

Jan 2025 – Jul 2025

Tech: Qwen 2.5 VLM, YOLOv8, PyTorch, Docker, FastAPI, MongoDB

Remote, Singapore

- Designed a real-time YOLOv8 pipeline for continuous RTSP streams (92.5 mAP50) and deployed a multimodal VLM (Qwen 2.5) pipeline for safety violation detection with structured JSON outputs.
- Engineered modular backend services (FastAPI, MongoDB) with RBAC and secure API management for scalable industrial AI pipelines; architected a multi-camera footfall analytics system using YOLOv8 and DeepSORT.

Dotworld Technologies Pvt Ltd | AI/ML Intern

Sep 2023 – Nov 2023

Tech: Generative Audio, CNN, Linux, MFCC

Coimbatore, India

- Benchmarked generative audio models (Tortoise, Bark) for latency and resource usage to guide deployment; developed a lightweight CNN-based wake word detection system using MFCC features, achieving 97% accuracy.

PROJECTS

Mini-vLLM: High-Performance LLM Inference Engine | CMU

Jun 2026 – Present

- Built a modular LLM inference engine from scratch (PyTorch, FastAPI) with a custom autoregressive decoding pipeline supporting greedy, temperature, top-k, and top-p sampling.
- Implemented KV cache, streaming inference, dynamic batching, prefix caching, and a simplified PagedAttention memory manager; benchmarked latency, throughput, TTFT, and GPU memory vs. Hugging Face.

Multi-Turn RL for SIMD Optimization (Qwen-2.5) | CMU

Sep 2025 – Dec 2025

- Architected an agentic multi-turn RL pipeline (GSPO) for AVX2 vectorization using verifiable rewards based on latency and correctness; standardized 287 kernels (TSVC/SIMDBench) into a dataset with execution harnesses.

Weakly Supervised Video Anomaly Detection (AGCN) | VIT Chennai

Jan 2025 – Apr 2025

- Built an adaptive graph convolutional network for video anomaly detection on UCF-Crime, achieving 92.15% AUC, surpassing state-of-the-art benchmarks using I3D features, attention, and K-max loss for segment-level supervision.

Systems-Centric DL for Predictive Maintenance (RUL Prediction) | CMU

Oct 2025 – Dec 2025

- Optimized Transformer inference with FlashAttention-2 and fused Triton kernels for RUL prediction on C-MAPSS, reaching 1.4M samples/s and 43x faster uncertainty estimation.

PUBLICATIONS

Abhinav Chaitanya R, Dharun Muthaiah N, et al. Autism Spectrum Disorder Detection using Attention-Based CNN and ML Classifiers. *Procedia Computer Science*, Vol 258, 2025. [\[Link\]](#)

S. S. Hemalatha, D. Muthaiah Nataraj, et al. Near-Infrared based Non-Invasive Blood Glucose Monitoring using Statistical Models. *2025 AI-Driven Smart Healthcare for Society 5.0*. [\[Link\]](#)